

FACULTY OF SCIENCE AND TECHNOLOGY

LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

PROGRAMME: BSSE/BSCS/BIT

YEAR/SEM: YEAR 2 SEM 1

COURSE CODE: BIT2115

PROBLEM NAME: JAVA VARIABLE TYPES

STUDENT DETAILS:

NAME OF STUDENT	Mulumeoderhwa Bahati Dieudonne
REG NUMBER	25/1512/BSCS-S

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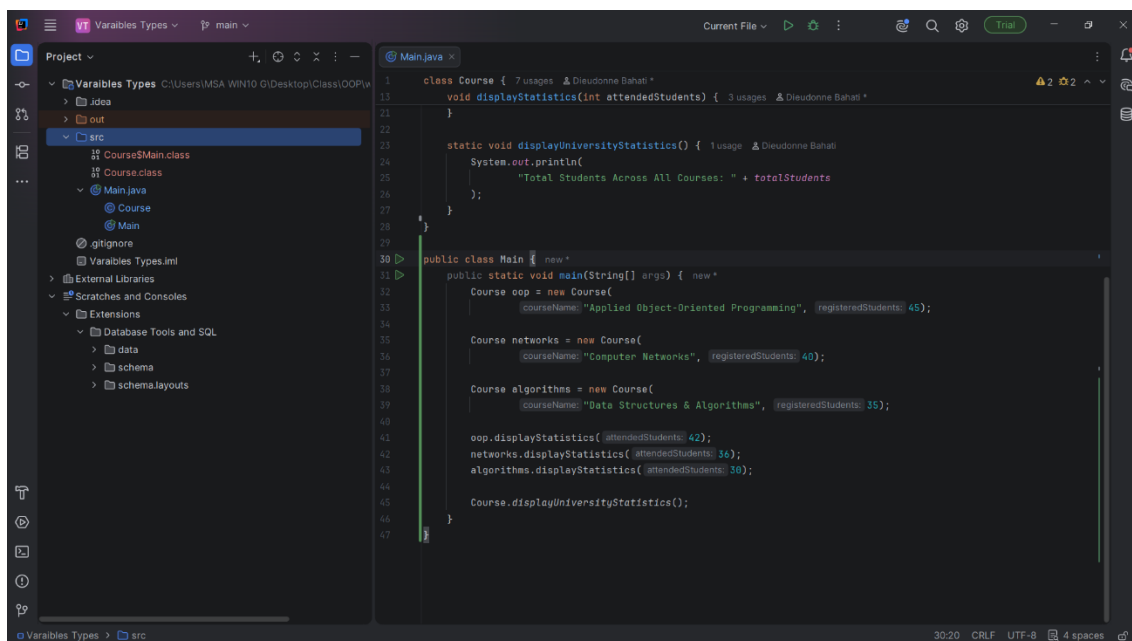
Section 1

Logic Building & Programming

Lines of code & explanations

Line Number	Explanation
#1	Declares the Course class.
#2–3	Declares instance variables for the course name and registered students; each object has its own values.
#4	Declares the static totalStudents variable shared by all Course objects.
#6–10	Constructor initializes each course object and adds its registered students to the shared total.
#12	Declares displayStatistics(), which displays statistics for one course.
#13–14	Declares a local variable, attendancePercentage, and calculates the attendance percentage temporarily.
#16–18	Displays the individual course name, registered students, and attendance percentage.
#21–23	Declares a static method that displays the university-wide total.
#26–28	Creates three Course objects with different course names and registration numbers.
#30–32	Calls displayStatistics() for each course using the object reference.
#34	Calls the static method to display the total across all courses.

Detailed Screenshots and caption descriptions



```

1  class Course {
2      String courseName;
3      int registeredStudents;
4      static int totalStudents;
5
6      Course(String courseName, int registeredStudents) {
7          this.courseName = courseName;
8          this.registeredStudents = registeredStudents;
9          totalStudents += registeredStudents;
10     }
11
12     void displayStatistics(int attendedStudents) {
13         double attendancePercentage = (double) attendedStudents / registeredStudents * 100;
14         System.out.println("Course: " + courseName + ", Registered Students: " + registeredStudents + ", Attendance Percentage: " + attendancePercentage + "%");
15     }
16
17     static void displayUniversityStatistics() {
18         System.out.println("Total Students Across All Courses: " + totalStudents);
19     }
20 }
21
22 public class Main {
23     public static void main(String[] args) {
24         Course eop = new Course("Applied Object-Oriented Programming", 45);
25
26         Course networks = new Course("Computer Networks", 40);
27
28         Course algorithms = new Course("Data Structures & Algorithms", 35);
29
30         eop.displayStatistics(42);
31         networks.displayStatistics(36);
32         algorithms.displayStatistics(30);
33
34         Course.displayUniversityStatistics();
35     }
36 }

```

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Figure 1:Class Main showing project structure

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Output and caption descriptions

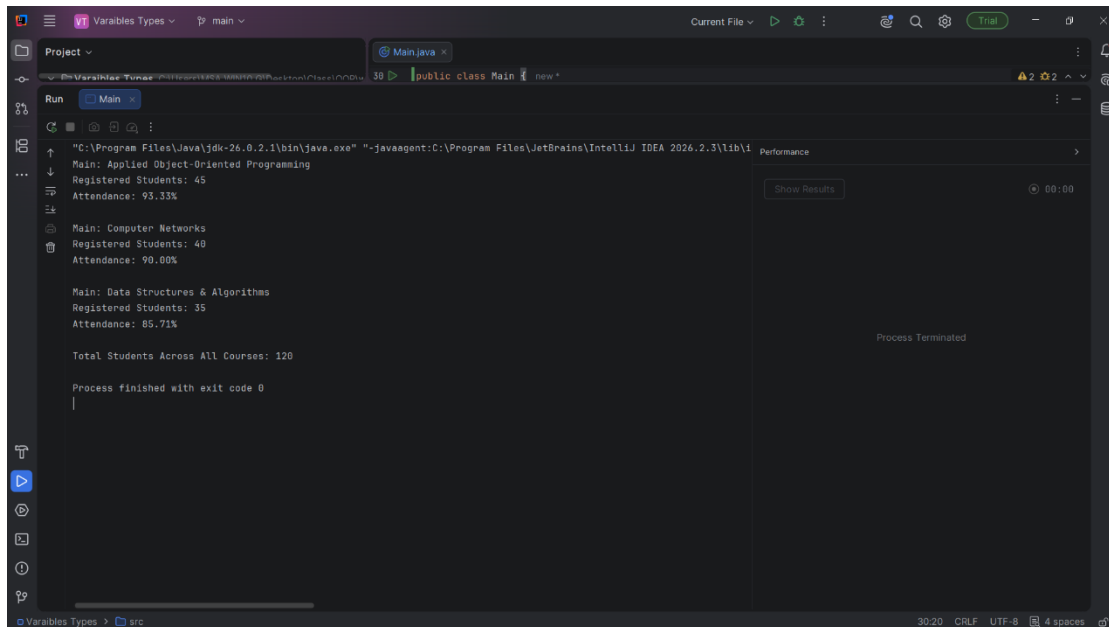


Figure 2: Output after running the Java enrolment program

Output Explanation

The program creates three course objects: Applied Object-Oriented Programming with 45 students, Computer Networks with 40 students, and Data Structures & Algorithms with 35 students. The displayStatistics method uses a local attendancePercentage variable for a temporary calculation. The static totalStudents variable is shared by all Course objects, so the university-wide total is 120 students. Each course object keeps its own courseName and registeredStudents values.

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Section 2

Version Control

Repository Overview

Repository Name	OOP
Repository URL	C:\Users\MSA WIN10 G\Desktop\Class\OOP\work\Varaibles Types
Branches	main

Commit #1 On Branch #1

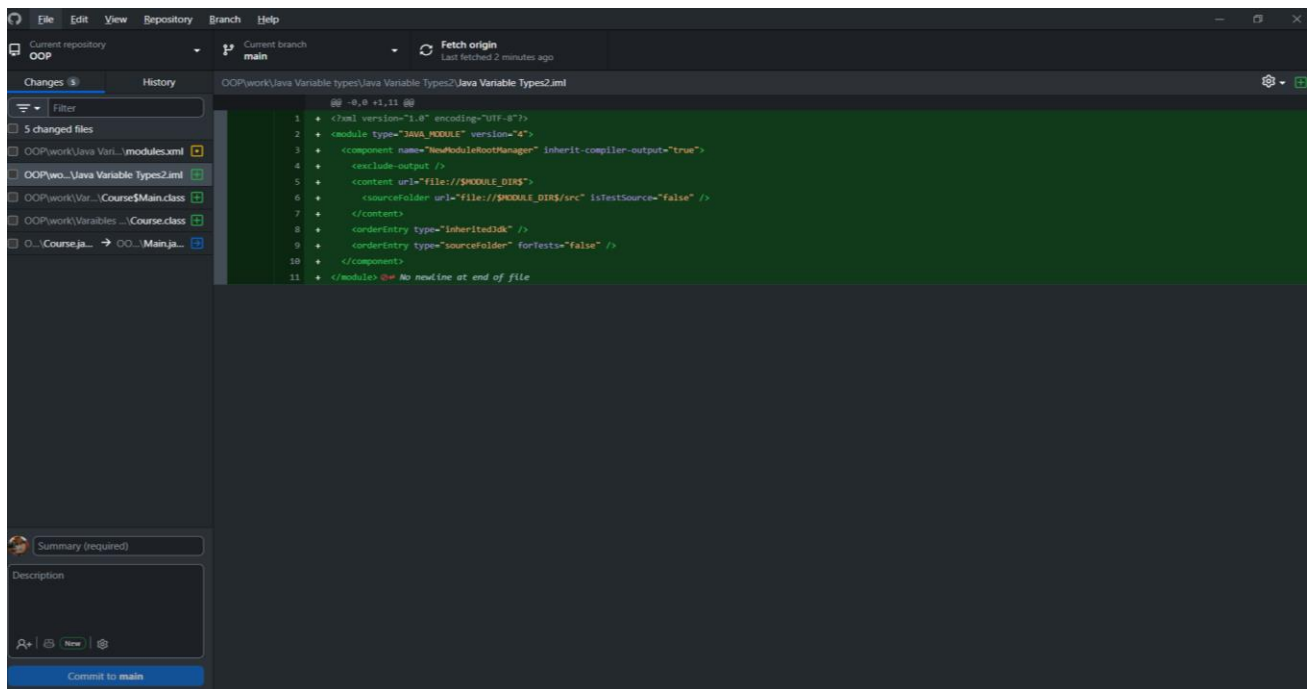


Figure 3: Initial commit on the main branch

Commit Summary

Commit Name	New: I created ne class Main instead of just course
Commit ID	c90da6f406660dbec91dc87a184f94cbe89cc9cd
Commit Description	New: I created ne class Main instead of just course
Branch Affected	main

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Section 3

What Have You Learned From The Task?

Skill Set	Description
Local Variables	Used inside a method for temporary calculations; attendancePercentage exists only while displayStatistics executes.
Instance Variables	Store data unique to each Course object, such as courseName and registeredStudents.
Static Variables	A single shared variable, totalStudents, keeps the total number of students across all Course objects.